

TRIX: Supplementary Material

Training-Free Reranking for Instruction-Following Retrieval with Exclusions

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A Full FollowIR Results

Table 1 provides the collection-level results underlying the aggregate comparison in Table ?? . Published instruction-trained values follow their source papers (???) .

Table 1: Collection-level FollowIR results. The two panels report retrieval effectiveness and p-MRR for the same systems; values are percentages ($\times 100$).

<i>(a) Retrieval effectiveness</i>				
Method	Core17	Robust04	News21	Score
<i>Instruction-trained retrievers</i>				
FollowIR-7B	20.0	24.8	29.6	24.8
Promptriever	21.6	28.3	28.5	26.1
InstEmb (Llama-3-8B)	24.0	29.2	32.3	28.5
<i>Matched frozen-backbone comparisons</i>				
BM25	10.8	13.8	21.0	15.2
+ <i>TRIX</i>	13.2	14.0	24.8	17.3
E5-Mistral	20.4	27.9	30.1	26.1
+ <i>TRIX</i>	24.4	26.7	32.8	28.0
BGE-large	16.8	19.4	21.4	19.2
+ <i>TRIX</i>	21.2	21.0	25.1	22.4
RepLLaMA	23.2	25.7	22.5	23.8
+ <i>TRIX</i>	25.6	27.9	31.8	28.4
<i>(b) Instruction sensitivity (p-MRR)</i>				
Method	Core17	Robust04	News21	Average
<i>Instruction-trained retrievers</i>				
FollowIR-7B	+16.5	+13.7	+6.3	+12.2
Promptriever	+15.4	+11.7	+6.4	+11.2
InstEmb (Llama-3-8B)	+22.4	+19.1	+5.4	+15.6
<i>Matched frozen-backbone comparisons</i>				
BM25	+3.7	−0.5	0.0	+1.1
+ <i>TRIX</i>	+18.4	+13.8	+20.3	+17.5
E5-Mistral	−0.3	−6.9	−0.9	−2.7
+ <i>TRIX</i>	+5.7	+19.6	+11.1	+12.1
BGE-large	−2.0	−5.2	+0.5	−2.2
+ <i>TRIX</i>	+4.3	+9.0	+6.3	+6.5
RepLLaMA	+1.1	−9.1	−1.8	−3.3
+ <i>TRIX</i>	+16.2	+14.1	+6.8	+12.3

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Table 2: Collection-level FollowIR comparison with query rewriting and expansion using RepLLaMA. The two panels report retrieval effectiveness and p-MRR; values are percentages ($\times 100$).

<i>(a) Retrieval effectiveness</i>				
Method	Core17	Robust04	News21	Score
RepLLaMA	23.2	25.7	22.5	23.8
HyDE	23.6	24.9	21.4	23.4
Query2Doc	25.9	26.1	24.9	25.6
ConvSearch-R1	22.3	25.9	22.0	23.4
TRIX	25.6	27.9	31.8	28.4
<i>(b) Instruction sensitivity (p-MRR)</i>				
Method	Core17	Robust04	News21	Average
RepLLaMA	+1.1	−9.1	−1.8	−3.3
HyDE	+6.5	−2.9	+0.7	+1.4
Query2Doc	+8.0	−7.5	−3.8	−1.1
ConvSearch-R1	+3.0	−2.5	+0.2	+0.2
TRIX	+16.2	+14.1	+6.8	+12.3

B Additional Ablations and Sensitivity

The main text isolates the positive view, residual exclusion signal, reward attenuation, and explicit penalty. The following results compare alternative residual estimators, examine sensitivity to the shared configuration, and report paired component differences.

B.1 Estimator and Normalization Choices

Table 3 compares the robust estimator, normalization, and residual construction used to obtain r .

Table 3: Secondary TRIX design choices on FollowIR. Each row changes one estimation or normalization choice while retaining the controlled candidate set and three evidence views.

Variant	Modification	Score	p-MRR
Full TRIX	None	28.4	+12.3
TRIX w/ OLS	Replace Huber regression	26.9	+10.6
TRIX w/ mean/std scaling	Replace median/MAD scaling	26.8	+8.5
TRIX w/o residual centering	Omit residual recentering	27.1	+10.8

B.2 Statistical Reliability

Because the controlled variants are paired at the query level, we report their paired differences in Table 4. All differences are computed per query and then averaged, so the reported Δ values reflect the mean per-query change rather than the difference of collection-level aggregates.

Table 4: Paired differences for the central component comparisons on FollowIR.

Matched comparison	Δ Score	Δ p-MRR
Positive-view scoring vs. full-query baseline	+1.7	+2.6
Residual exclusion subtraction vs. raw exclusion subtraction	+2.6	−2.2
Full TRIX vs. TRIX w/o reward attenuation	+2.1	+1.5
Full TRIX vs. TRIX w/o exclusion penalty	+2.0	+3.4

Table 5: Sensitivity and efficiency. Score and p-MRR are averaged over FollowIR; latency is measured per query under a fixed hardware setting.

Analysis	Setting	Score	p-MRR
$\lambda = 0.5$	default	28.4	+12.3
$\lambda = 1.0$		27.6	+10.8
$\lambda = 1.5$		26.9	+9.2
$\lambda = 2.0$		25.8	+7.5
$\tau = 0.1$	default	28.4	+12.3
$\tau = 0.2$		28.2	+11.9
$\tau = 0.5$		27.8	+10.4
$\tau = 1.0$		27.1	+8.7
$K = 10$	depth	26.3	+18.1
$K = 50$		27.9	+20.8
$K = 100$	default	28.5	+22.5
$K = 200$		28.4	+22.1
Decomp.	LLM call	0.3 s/q	
TRIX rerank	Emb. cache	5 ms/q	

Table 5 reports whether the shared scoring hyperparameters and candidate depth change the main conclusion, together with the end-to-end overhead required by decomposition and reranking.

C Additional Exclusion-Benchmark Results

Table ?? (main text, Section ??) reports the complete TRIX results across NegConstraint and ExcluIR. Table 6 examines whether its performance depends on using the per-query regression residual instead of raw exclusion similarity. The same matched comparison is applied to both benchmarks. All rows use the same frozen retriever, candidate set, decomposition, query templates, document embeddings, and shared λ, τ ; the raw and residual correction rows differ only in the exclusion signal supplied to the same TRIX scoring function.

Under the same TRIX scoring rule, the regression residual provides a stronger effectiveness–exclusion profile than raw

Table 6: Matched exclusion-signal ablation across exclusion benchmarks with BGE-large-en-v1.5.

Variant	NegConstraint		ExcluIR	
	MAP	nDCG@10	MRR	RR
Full-query baseline	63.8	71.9	73.4	56.0
TRIX w/ positive view only	76.0	81.8	75.0	58.0
TRIX w/ raw exclusion score	76.6	81.8	76.0	60.0
Full TRIX	76.6	81.8	78.2	82.2

similarity on both benchmarks. The residual ranking signal therefore remains effective under both evaluation protocols.

D Qualitative Analysis

The qualitative analysis contains one representative success and one weak-exclusion-signal failure from FollowIR (Core17) with RepLLaMA. Each case reports the benchmark, query ID, complete request, decomposed views, candidate scores, and the ranking change. The success case is the query with the largest single-document rank drop; the failure case is the query with the smallest rank change among those carrying an exclusion instruction.

Table 7: Representative TRIX cases from FollowIR Core17 with RepLLaMA. Each case reports the operational views, score-space evidence, and ranking change under the complete scoring rule.

Outcome	Request and views	Candidate evidence	Ranking effect and interpretation
Success	Benchmark: FollowIR (Core17), QID 355 Query: “Identify documents discussing the development and application of space-borne ocean remote sensing” Exclusion: land-bound sciences, agriculture, forestry, marketing, temperature Views: Q_{full} (full query), Q_{pos} (ocean remote sensing in oceanography, seabed prospecting, marine sciences), Q_{neg} (land-bound sciences, agriculture, forestry, marketing, temperature)	Original top-1 doc ($s_{\text{full}} = 0.725$) scores high on both s_{pos} and s_{neg} channels (agriculture/forestry content). Residual r separates the exclusion signal from shared marine-science relevance. After TRIX: top-1 score rises to 1.907 for ocean-only doc; land-bound doc drops to rank 580.	Land-bound/agriculture documents drop from top-10 to rank > 500. Ocean remote-sensing documents rise to top-5. Interpretation: Residual penalty correctly identifies exclusion signal when exclusion and target content are topically distinct.
Failure	Benchmark: FollowIR (Core17), QID 310 Query: “Evidence that radio waves from radio towers or car (mobile) phones affect brain cancer occurrence” Exclusion: leukemia Views: Q_{full} (full query), Q_{pos} (radio waves and brain cancer incidence), Q_{neg} (leukemia)	Original top-1 doc ($s_{\text{full}} = 0.795$) discusses brain cancer broadly. s_{neg} channel for “leukemia” is low because leukemia-specific mentions are rare in the candidate set, yielding near-zero residual r for the top document. After TRIX: top-1 score rises to 1.689 but ranking is unchanged.	Top-5 ranking is unchanged; excluded-topic documents remain in place. Interpretation: When the exclusion term rarely co-occurs with the target in the candidate set, the residual is too small to trigger a penalty—the failure mode is low negative-channel signal rather than semantic entanglement.

E Decomposition and Reproducibility Details

The instruction decomposer is the frozen Qwen3-4B model (?). It receives the complete user request and returns structured affirmative and exclusion content in one generation. The full view Q_{full} is the original request itself, used unchanged for candidate generation; Q_{pos} retains the topic and affirmative conditions while removing exclusion content, and Q_{neg} contains the excluded content, or [NONE] when no exclusion is present. We use the fixed TSC_BALANCED_t01 prompt with temperature 0.1, top- $p = 0.9$, at most 512 new tokens, and thinking disabled. Malformed or missing fields are mapped to the corresponding empty view. The system prompt is:

```
You are an expert IR query optimizer for Dense
Vector Models. Decouple "Original Query" and
"Instruction" into Q_plus and Q_minus.
```

```
Q_plus rules: (1) Fluent natural language phrase,
not a keyword list. (2) Semantic topic only;
remove format noise ("articles", "documents",
"reports"). (3) Remove conversational connectors
("such as", "including"). (4) Sharp, coherent.
```

```
Q_minus rules: (1) Extract only critical
high-level exclusion concepts. (2) Concise
natural phrasing. (3) If no exclusion,
output "[NONE]".
```

```
No-negation rule: DENSE RETRIEVERS IGNORE LOGIC
WORDS. NEVER use negation in Q_minus ("no",
"not", "non-", "without"). Convert negations to
AFFIRMATIVE entities:
```

```
BAD: "non-United Kingdom"
-> GOOD: "United States, France, global"
BAD: "violence outside Ireland"
-> GOOD: "violence in England, international"
Output ONLY exact entities to exclude.
```

Output JSON:

```
{
  "Reasoning_Steps": "1. Core topic. 2. Filter
    noise. 3. Exclusions (negations->affirm).",
  "Q_plus": "Fluent natural language query",
  "Q_minus": "Core exclusions or [NONE]"
}
```

The user prompt template is:

```
Decouple into Q_plus and Q_minus:
Query: "{query}"
Instruction: "{instruction}"
```

All document representations are computed once with the base retriever’s document template, and all three query views use its query template. For numerical stability, Huber residuals and channel scores use a 10^{-6} floor in their median/MAD normalization. The regression is fit independently on each candidate set and uses no relevance labels. The scoring hyperparameters λ and τ are fixed before evaluation and shared across datasets and retrievers; no query-specific tuned coefficients or metric-based tuning are introduced. The released configuration records these constants and the candidate-set depth for each experiment.

References

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